PREDICTING NATIVE INTERVENTION OUTCOMES IN SPECULATIVE DECODING

Anonymous Authors[1]

ABSTRACT

A reference mismatch rejects an execution path but does not describe its later output or predict a controlled intervention. We test frozen executable execution-path descriptions against complete native generated-ID arrays and endpoints before and after stage-specific changes. On one B570 stack, draft-free predictions match both conditions on 24 exposed tasks, and a prefill intervention restores all four diagnostic deviations without new control deviations. On one RTX stack, proposal-conditioned predictions match both conditions on 43 tasks, yet a target-continuation intervention fails its reference-recovery and no-new-regression goal: one task recovers and one newly deviates, leaving aggregate agreement unchanged. These distinct finite tests establish intervention-prediction closure, not universal repair, functional correctness or a unique stage cause. Separate cost evidence finds qualified RTX serialization slower, conditional 358H overhead-removal ceilings below parity, and M4 Guard inferior to fixed speculation. Complete prediction, paired reference effects and realized cost support different system decisions.

1 INTRODUCTION

Speculative decoding uses cheap proposals to amortize target work (Leviathan et al., 2023). Its algorithmic output guarantee does not by itself certify a concrete numerical execution path. Prior studies document batch-shape sensitivity, correctness gates and slowdowns (Chordiya, 2026; Yuan et al., 2025; Zhang et al., 2026). A failed reference comparison rejects a path as a strict replacement, but says little about its complete divergent output or what a controlled change will do to that output.

We study frozen executable execution-path descriptions: programs that compute complete generated-ID predictions before native execution. They are not learned models. Prediction/native equality asks whether a description forecasts the realized path; native/reference equality asks whether that path preserves a specified greedy reference. A controlled intervention can satisfy the first comparison while failing the second. Even a reference-compatible path needs an independent cost test before it can support an optimization claim.

On a B570 Vulkan stack, a draft-free target-only description predicts the native serialized path on 24 tasks. A later controlled intervention changes target prefill from UB1 to UB128, retains UB1 continuation and drafting, and transfers the first pending sample and state at the frozen boundary. Predictions match native outputs before and after the change; all four diagnostic tasks recover complete reference agreement and all 20 stable controls remain compatible. This is successful restoration on an exposed cohort, not identification of a unique prefill cause.

On an RTX CUDA stack, a different description uses proposals and computes target decisions through separately orchestrated APIs on shared numerical routines. A frozen target-continuation UB2-to-UB1 intervention preserves UB2 prefill, UB2 drafting and caller grouping. Predictions match complete native outputs in both conditions on 19 exposed HumanEval tasks and 24 exposure-audited MBPP tasks. The candidate nevertheless fails its reference-recovery and no-new-regression goal. One HumanEval task recovers, another remains divergent, and a previously matching task acquires a deviation. Aggregate reference agreement is unchanged and conceals these opposed paired effects.

These studies test distinct descriptions, interventions, stacks and cohorts. They establish finite prospective intervention-prediction closure, not transfer of one predictor or repair across hardware. Their practical use is to test an intervention against its realized native path, including a candidate that should be rejected. The contribution does not rest on aggregate cancellation, which is elementary paired accounting; it rests on the frozen information contracts, complete-output tests and controlled native comparisons that make the cancellation interpretable.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the Machine Learning and Systems (MLSys) Conference. Do not distribute.

We contribute (1) prospective complete-output descriptions validated before and after two controlled interventions; (2) paired reference comparisons that distinguish restoration, persistent deviation and new deviation while preserving the RTX candidate's failed target; and (3) a separate qualification/cost decision framework grounded in matched RTX timing, conditional 358H budgets and held-out M4 control. RTX serialization qualifies yet is slower; 358H overhead removal cannot reach parity at fixed target work and progress; M4 Guard loses to fixed speculation. Prediction and restoration therefore support diagnosis, while actual cost determines optimization eligibility. We introduce neither a new decoder nor a production repair or superior adaptive policy.

2 Questions, Output Contract and Cost

We distinguish complete-output prediction, reference compatibility, paired intervention outcome and realized cost. A program may predict a changed path correctly while that path remains incompatible, and compatibility alone does not establish a cost benefit. Acceptance rate, isolated draft throughput and target-forward batching answer narrower questions.

2.1 Prediction and reference compatibility

Let k be maximum draft width, UB the target micro-batch capacity (n_{ubatch}), and EOG a vocabulary end-of-generation ID. A greedy speculative round verifies proposals, commits the agreeing prefix and a correction or bonus when budget permits, then removes inconsistent suffix state. Our strict output contract compares complete committed generated IDs with the matched target-only greedy reference under the frozen model/runtime/configuration. A failed pair blocks a workload-global lossless optimization claim; a small passing probe establishes eligibility for a larger gate. Capped ID equality does not measure functional HumanEval correctness or pass@k.

Historical text-path agreement (H) compares arrays reconstructed from CLI text; direct-ID qualification (Q) compares captured IDs with the study reference. For the intervention studies, write R for that reference, P_0, P_1 for sealed baseline/intervention predictions and N_0, N_1 for the corresponding native outcomes. A complete trajectory here is the generated-ID array and endpoint within the measured boundary, not every internal state, logit, kernel or execution event. Prediction equality $P_j = N_j$ and reference compatibility $N_j = R$ have separate denominators.

A paired task is restored when $N_0 \neq R$ and $N_1 = R$; it has a new deviation when $N_0 = R$ and $N_1 \neq R$. Persistent deviation means both fail the reference comparison. It is an output no-op only if the complete N_0, N_1 arrays and endpoints also agree. These labels concern specified reference compatibility, not program functionality or HumanEval pass rate. Each task has three stability observations, not three independent tasks.

2.2 Execution descriptions as testable candidates

An execution-path model is a frozen executable description with a stated input contract. B570's intervention predictions are draft-free target-only programs; RTX's are proposal-conditioned application/API orchestrations sharing pinned numerical routines with native. They are different programs with different inputs, not a single transferred predictor. Table 1 makes those differences explicit; historical candidate contracts remain in the appendix. UB is micro-batch capacity, not observed occupied graph shape or the sequence of forwards. Matching capacities need not match numerical paths.

2.3 Complete realized cost

Let T_b be matched target-only cost per output token, T_d^{real} complete realized draft-subsystem time including interface replay, $T_{\mathrm{target,round}}^{\mathrm{real}}$ complete target work, and T_x residual synchronization, transfer and bookkeeping outside those regions. Let C count all committed output per round, including correction/bonus. Then

$$S = \frac{T_b C}{T_d^{\mathrm{real}} + T_{\mathrm{target,round}}^{\mathrm{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\mathrm{real}} + R_{\mathrm{target}} + X}. \tag{2}$$

The normalized denominator gives break-even $C > D_{\mathrm{real}} + R_{\mathrm{target}} + X$. These are accounting identities, not new profitability theory (Kim et al., 2026). Total-time ratios need not equal prompt geometric means. Incompatible experiments cannot isolate an unmeasured component by subtraction. The historical target-forward proxy V_k^{micro} omits drafting and integration; its original cells and separate fresh qualification gates remain in Appendix F.

Primary results retain their frozen questions and gates; post-study results are identified separately. Timing on blocked paths remains diagnostic, and a later Q gate does not certify historical H timing.

3 Study Design and Evidence Scope

3.1 Frozen systems and complete endpoints

Each study has its own protocol, tasks and comparator, stated beside its results and mapped in the historical appendix and the separate intervention appendix. The target/draft pair is Qwen3-8B/0.6B Q8_0, except the 358H adapter's symmetric INT4 OpenVINO draft. Final compar-

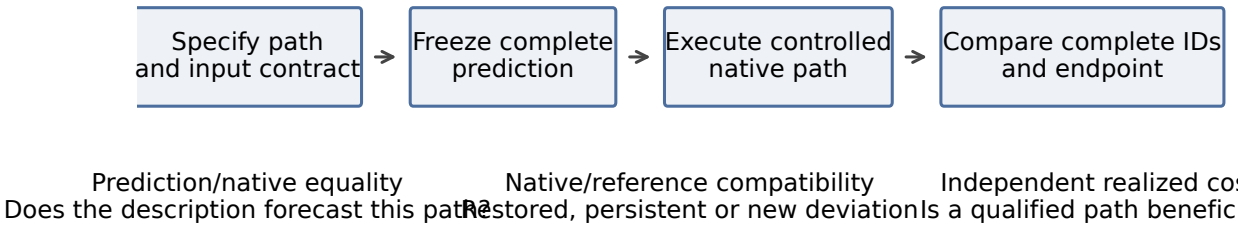


Figure 1. Separate decisions for each frozen native configuration. A description predicts the complete generated-ID array and endpoint; native execution is then compared with that prediction and its own specified reference. A changed path may restore, retain or introduce reference deviation. Compatibility supplies eligibility for an independent cost test, not a measured gain.

isons process one request at a time. Historical settings are in Appendix K and RTX V4 in Appendix Q; hardware, backend, OS and quantization jointly vary, preventing hardware-only causal comparisons. RTX's original direct-ID OS build is 26200; follow-ups use 26300. A driver version alone does not identify driver bytes.

"Complete" retains each protocol's generated-ID and endpoint rules: prompt IDs and uncommitted proposals are excluded; an earlier terminal EOG is inclusive, otherwise the 128-ID cap applies. Actual stop markers and raw overshoot remain evidence rather than being silently truncated by a common-prefix test. Intervention comparisons include both the complete array and endpoint. Valid results are selected from each result manifest's complete attempt; failed or interrupted attempts are retained separately and are never concatenated.

The B570 intervention uses a previously exposed 24-task diagnostic/control cohort. RTX uses the old 19-task cohort and a separately frozen 24-task MBPP cohort; its candidate, exposure exclusions and static task order precede new-task execution. Frozen prediction arrays precede native runs. RTX additionally records remote readback before native; B570 uses local seals. Neither provides independently trusted absolute experimental time. Prospective prediction is not equivalent to formal independent preregistration, and the candidate's earlier exposure is recorded. No tasks are added or dropped after observing these results. Three repeats assess within-task stability; finite exact counts receive no pseudo-independent binomial intervals.

The offline intervention review restores the hash-bound evidence archive and independently parses selected result/raw/trace records. It checks prediction/native and native/reference arrays and endpoints, repeat stability, paired transitions, handoff witnesses, shams, graph occupancy and effective settings. Opaque sampler state cannot be serialized: clone behavior and checked surrounding state do not establish byte equality of opaque sampler bytes. Omitted weights/runtime binaries remain identified but are not independently rehashed or executed by this review. The author evidence packet records exact review coverage and source/hash bindings.

Historical M4 includes a representative 24-prompt SpecBench sample (Xia et al., 2024) and a separate preregistered 30-task HumanEval study (Chen et al., 2021); they are background, not replications of the later runtime study. The latter retains its own calibration, qualification and held-out splits in Section 6. All studies keep their original stop rules and comparators.

3.2 Cost estimators and conditional uncertainty

Later RTX cost studies take each condition's median of three times per prompt, then reference/candidate ratios and their prompt GM. V4 uses 10,000 paired-row bootstrap draws, seed 2026100204, frozen task order and linear .025/.975 percentiles. This resamples the empirical distribution of the selected prompt rows, conditional on the exposure-audited cohort, observed condition medians and frozen configuration. It does not account for exposure selection, predict a HumanEval population failure rate or cover a new thermal/session distribution. Repetitions are not independent bootstrap units. Cost interpretation requires every reference-ID pair and stable reference repeats; blocked timing remains diagnostic.

The 358H prompt estimator takes the median of three paired ratios before the prompt GM; total-time accounting is separate. M4 runtime uses condition medians and its own frozen random streams. Historical H estimators remain in Appendix B. All intervals omit full session and recalibration uncertainty.

Timing excludes loading and initial prefill. The RTX V4 timer begins after synchronized target/draft prefill and before per-request sampler/state work, ending at slot release.

The M4 and 358H boundaries differ and are stated with their results. M4 package energy excludes separate DRAM, B570 uses adapter-board energy, and RTX mixed domains do not form a common system boundary. There is no end-to-end 358H Phase-3 or runtime energy result.

3.3 Offline reproducibility

The anonymous index recomputes retained arrays, timer intervals and summaries; the author packet supplies source/hash bindings and raw-review coverage. Appendix O maps historical entries and Q the V4 capabilities. Verification runs no models and does not authenticate omitted assets or absolute process time.

Existing-record analysis. Score/margin descriptors, trajectory extents, fixed-mode opportunity and conditional cost bounds are calculated after aggregate outcomes are known. Extent/budget analyses retain original rows and comparators without adding samples or intervals. V4 latency summaries describe 19 prompt medians per condition without changing the paired GM/CI. Appendix P gives the derivations.

4 Prediction Before and After Controlled Intervention

4.1 Distinct interventions and input contracts

The two studies ask whether frozen descriptions predict complete native outputs before and after a controlled change. They do not compare hardware under a common intervention. Table 1 aligns their configurations and information boundaries. The separate intervention appendix maps the abstract P_j notation to original study labels and retains every task's paired outcome. Earlier target-only candidate tests and sampled forks remain in the appendix with their original questions.

B570 changes target prefill UB1-to-UB128 and includes the protocol's first-pending sample/state handoff; target continuation and draft remain UB1. RTX changes target post-prefill continuation UB2-to-UB1, keeping target prefill, draft UB2, $k = 2$ and caller grouping fixed. Both use same-boundary unchanged-capacity shams. Native paths receive prompt, assets and fixed parameters, use their own live draft, and receive no prediction tape, reference IDs or future native answers. The RTX description's separately orchestrated proposal/target workers share pinned numerical routines with native; this is not an independent numerical implementation.

4.2 B570: restoration on the tested exposed cohort

The 24-task intervention cohort contains four previously exposed diagnostic tasks and 20 stable controls. Baseline prediction/native equality is 72/72 and intervention equality is 72/72. Reference agreement changes from 60/72 to 72/72: HumanEval/134, /42, /145 and /138 recover in all three observations, with no new deviation among the controls. Prediction and native execution are separately checked against the complete capped arrays, not their common prefixes.

The intervention supplies a successful controlled native restoration on this configuration. It does not isolate a pure KV factor: the first pending token is sampled and accepted once in the source, then KV, pending logits, positions and a post-accept sampler clone are transferred. The exposed tasks are diagnostic validation, not a new unexposed generalization estimate. The earlier draft-free predictor still establishes a simpler reproducer; the older sampled fork's changed scores, stable immediate decisions and confounded delayed divergence retain their original causal limits.

4.3 RTX: prediction closes while the recovery target fails

The fixed candidate is evaluated on 19 exposed HumanEval tasks and 24 statically selected MBPP tasks with no known prior execution exposure within the audited project records. This does not imply training novelty or a random benchmark sample. For each cohort, both baseline and intervention predictions match every complete native array and endpoint: 57/57 per condition on HumanEval and 72/72 per condition on MBPP. All 43 tasks are repeat-stable. The two prediction comparisons are not 258 independent tasks.

Figure 2 shows the paired reference effects. HumanEval/15 changes from mismatch to match; /18 remains divergent; /25 changes from match to mismatch. Thus HumanEval reference agreement remains 51/57 arrays, or 17/19 tasks, before and after. On MBPP, 22 tasks remain matching and /463 and /304 remain divergent, giving 66/72 arrays in each condition. Prediction closure therefore coexists with failure of the predefined reference-recovery plus no-new-regression goal. That goal is not replaced by a post-study definition of success.

Persistent reference deviation alone says nothing about output no-op. HumanEval/18 and MBPP/463 retain identical complete native outputs and EOS endpoints. MBPP/304 changes its 47-ID output at zero-based index 33 (488 to 481), yet remains divergent. The appendix provides full endpoint and first-difference witnesses. Across RTX the native/reference total is 117/129 in each condition: 43 tasks with three stable observations, not a population success rate.

Target-continuation UB2-to-UB1 alone did not reproduce historical global-UB1 reference agreement under this controlled configuration. The historical global option also changed prefill and draft, and historical native defaults were effectively 12 threads despite a 24-thread description; this

Table 1. Distinct controlled studies. UB denotes capacity, not observed graph occupancy. Counts are complete arrays/endpoints over three stable observations per task; cohorts are not pooled into a general success rate. Neither intervention has a performance or energy measurement.

Contract	B570 / Vulkan	RTX / CUDA
Cohort	24 exposed tasks: 4 diagnostic, 20 stable controls	19 exposed HumanEval; 24 exposure-audited MBPP
Baseline	Target prefill/continuation UB1, draft UB1	Target prefill/continuation UB2, draft UB2
Changed stage	Target prefill UB1→UB128 plus protocol first-pending sample/state handoff	Target continuation UB2→UB1 after UB2 prefill; first-pending sample/state handoff
Fixed path	Target continuation UB1, draft UB1, $k = 2$	Target prefill UB2, draft UB2, $k = 2$, caller grouping
Prediction inputs	Target-only prompt/assets/configuration; no draft or proposals	Own proposal worker and target worker; shared pinned numerical routines
Native inputs	Own live draft; no sealed prediction inputs	Own live draft; no H tapes/reference IDs
Sham / seal	Same-boundary UB1→UB1 / local seals	Same-boundary UB2→UB2 / remote readback before native
P_0/N_0; P_1/N_1	72/72; 72/72	57/57; 57/57 (old), 72/72; 72/72 (new)
$N_0/R \to N_1/R$	60/72→72/72	51/57→51/57; 66/72→66/72

study explicitly uses 24. Its effective G0 contains 75 newly checked native arrays at 24 threads and 45 unchanged H arrays reused, not 120 new 24-thread outputs. These differences prevent attributing the contrast to a necessary prefill factor, drafting factor or interaction. HumanEval/15's restoration also prevents labeling every target-continuation intervention ineffective.

5 Qualification and Optimization Limits

5.1 RTX: matched qualification and complete cost

V4 B uses A's 19 tasks but 171 fresh observations: target-only UB1, batched $k = 2$ UB2 and serialized $k = 2$ UB1. One frozen timer executable, model/runtime, input, endpoint and balanced order schedule cover all three paths; each condition occupies each order position 19 times. The global native micro-batch option also propagates to draft, making this a whole-path configuration comparison, not an isolated target intervention.

The fresh ID gate blocks batching at 51/57. Serialization qualifies at 57/57, but its prompt-median target/candidate GM is 0.677× [0.662, 0.692], with 0/19 prompts faster (Table 2). Reciprocal relative GM latency is 1.476×. The experiment excludes the measured serialized configuration as a latency optimization and establishes no fast qualified alternative. Blocked batching supplies no admissible speedup or CI.

Absolute latency descriptors show the scale behind the ratios (Table 3). Median across the 19 prompt medians is 1.607s for target and 2.332s for serialization; the full ranges are 0.532–2.920s and 0.843–4.102s. These post-study distributions do not estimate the paired GM by dividing their medians. Appendix Q retains every prompt-condition median and the unchanged primary estimator. The earlier two-path 30-prompt result, 0.646× [0.630, 0.660], uses a different cohort and has no batched condition; the two estimates do not form an optimization trend.

The native timer charges per-request sampler/state work after synchronized prefill, drafting, target validation, acceptance/correction/bonus, KV rollback, synchronization and

Table 2. RTX V4 matched qualification and cost: 171 fresh runs on 19 prompts × three × three paths. Target/serialized use UB1, batching UB2; speculation uses $k = 2$. Speedup = target/candidate latency. Only the blocked batched row has null speedup/CI. Serialization is faster on 0/19 prompts.

Path	ID gate	Speedup [95% CI]
Target	Reference	1.000
Batched	51/57 blocked	—
Serialized	57/57 qualified	0.677 [0.662, 0.692]

Table 3. RTX V4 absolute prompt-median post-prefill latency (s): 19 medians per condition, each from three runs. Post-study spread, not uncertainty. Batched times are diagnostic; these distribution medians do not estimate the paired speedup.

Condition	Min	Q25	Median	Q75	Max
Target	0.532	1.441	1.607	2.396	2.920
Batched	0.641	1.730	1.910	2.953	3.248
Serialized	0.843	2.074	2.332	3.611	4.102

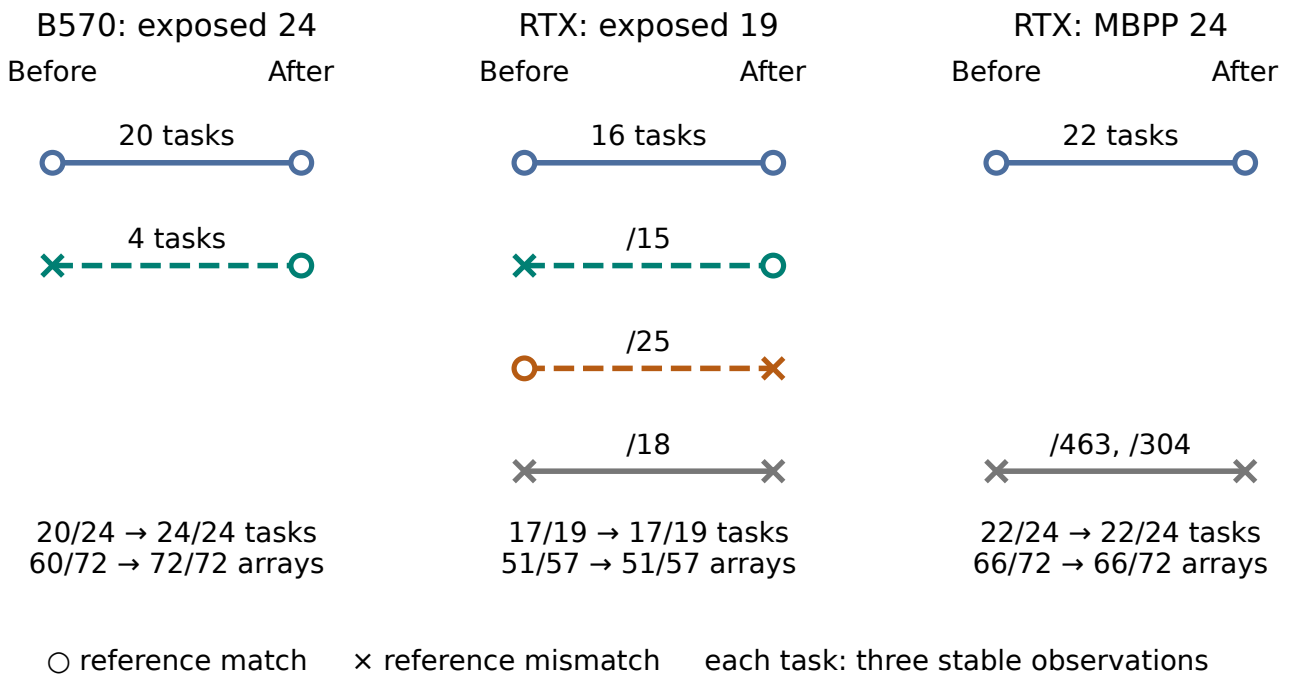


Figure 2. Task-level paired reference effects. B570 restores four diagnostic tasks with no new deviation among 20 controls. RTX HumanEval restores /15 and newly deviates on /25, leaving 17/19 tasks (51/57 repeated arrays) matching in both conditions; MBPP is shown separately. Each task has three stable observations. Equality compares the full protocol-selected generated-ID array and endpoint (inclusive earlier EOG or cap, with raw overshoot retained); mismatch is not functional failure. Equal aggregate agreement can conceal paired restoration and new deviation.

related logging/response through slot release. Loading, prefill and teardown are excluded. Observer controls check output stability, while clock probes bound clock-call overhead only; condition-dependent debug logging need not cancel. Thus the conclusion concerns instrumented post-prefill request latency, not logging-free deployment, throughput or energy. The bootstrap conditions on the selected rows and observed medians as stated in Section 3.

5.2 358H: a bound on draft-only optimization

The closed 358H Phase-3 runner directly qualifies CPU and NPU drafting at 90/90 pairs per path, yet both are slower than target-only: CPU 0.734× [0.693, 0.771], NPU 0.352× [0.333, 0.369], with zero of 30 prompts faster. Isolated NPU drafting is faster than CPU at tested contexts, but the integrated synchronous OpenVINO region combines full-prefix replay/reconstruction and proposal. Its 58.35% share cannot be separated into the two costs (Figure 3). This is an interface result, not a general NPU limitation.

Measured committed progress is $C = 2.4631$ CPU and 2.4584 NPU, including correction/bonus. Complete target work consumes 87.34% of CPU-path time; residual orchestration is 0.04% CPU and 0.02% NPU. Processed-total accounting gives $(D_{\text{real}}, R_{\text{target}}, X) = (0.4304, 2.9767, 0.00124)$ CPU and $(4.1251, 2.9435, 0.00130)$ NPU. Their 0.7227× and 0.3477× total-time ratios differ from prompt GMs.

With baseline, target work and progress fixed, eliminating draft/residual gives pooled-total ceilings $C/R_{\text{target}} =$ 0.8275× CPU and 0.8352× NPU. The original prompt estimator has its own bound. Let B_{ij} and T_{ij} be matched baseline and complete target time for prompt i, repetition j. Nonnegative remaining cost and monotonicity of the median and GM give

$$\mathrm{GM}_i \,\mathrm{median}_j\, S'_{ij} \le \mathrm{GM}_i \,\mathrm{median}_j (B_{ij}/T_{ij}). \quad (3)$$

All 90 rows per path yield ceilings 0.8372× CPU and 0.8448× NPU. Thus overhead-only savings cannot reach parity under either estimator. Even with other cost zero, a uniform target-work reduction must be at least 16.28%/15.52% to make the prompt-GM ceiling reach one. These are post-study necessary budgets, not achieved savings; changed progress or target execution requires new qualification and measurement. Appendix P gives full precision and per-prompt limits. The persistent-state/wider-target candidate stops at qualification, with no performance or energy result.

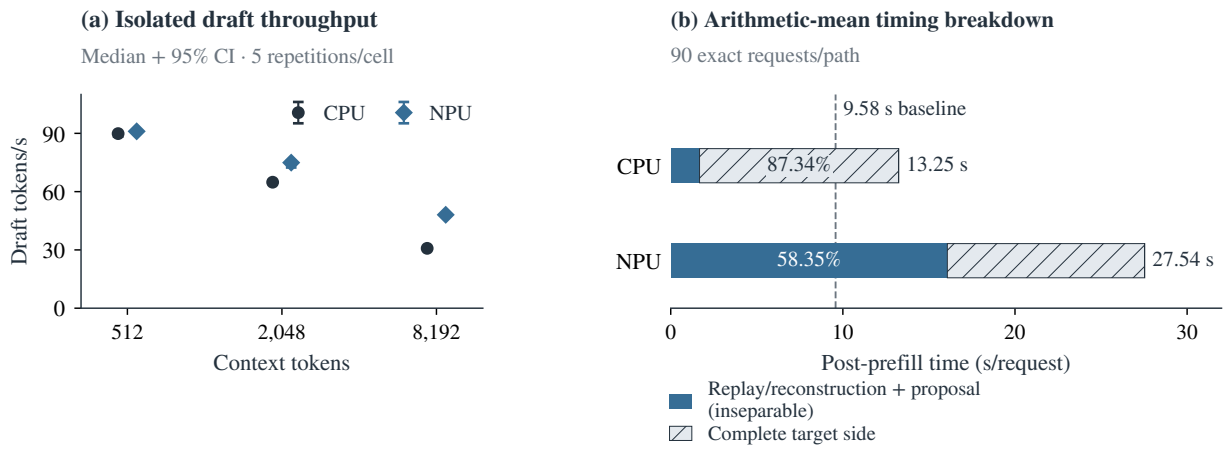


Figure 3. **Isolated draft advantage and complete 358H cost.** A: five-repeat steady-state draft throughput with repetition-bootstrap intervals. B: arithmetic mean post-prefill seconds over 90 exact observations per path, against the shared baseline. NPU's 58.35% region combines full-prefix replay/reconstruction and proposal; it is inseparable. CPU complete target work is 87.34%. These bars differ from the prompt-GM speedups in Section 5.2; persistence savings are unmeasured.

CPU restore/reconcile correctness could not be established under the tested state representation; the NPU state-capable path was unavailable with the frozen dynamic-shape IR. Separate correctness-only controls reproduce UB1/UB2 greedy divergence on the target even with exact matching prefixes, and proposal replay reproduces speculative divergence without live NPU draft inference. These controls identify no unique floating-point operation. The implementation motivated by cost remains blocked by output compatibility; neither device is proved incapable of retaining state.

6 Runtime Control under Qualified Fixed Modes

The separate M4 runtime study tests whether continuation control improves a qualified speculative path. It uses its own frozen policy, direct-ID gate and cost boundary; historical M4 timing is background rather than its qualification evidence.

6.1 A prospective cost guard on M4

The M4 experiment asks whether a frozen rule based on already-realized complete cost can control continuation while preserving exact output and profitable execution against target-only. It does not assume that adding control improves on fixed speculation. We retain the Qwen3-8B/0.6B Q8_0 pair on llama.cpp Metal, fixed $k = 2$, greedy generation, and a 128-token cap. Guard begins on the same speculative path as Always-Spec and can fall back permanently to target-only within a request. It changes neither proposal width nor device placement.

At each eligible completed round, Guard divides all elapsed charged generation cost by committed output progress C. This count includes the initial target seed and committed correction or bonus tokens, but excludes rejected proposals. Guard continues if this cumulative cost per token is at most $(1 - m)R_{\text{ref}}$, a target-only reference frozen from calibration; otherwise it falls back. If a decision is reached with zero committed progress, Guard also falls back. Before each control decision, invalid candidate suffix state is removed from the target and draft contexts, preserving committed history for target-only continuation. The enclosing post-prefill clock includes generation-loop initialization, seed-token sampling, drafting, target work, synchronization, state reconciliation, control, and transition costs. Model loading, sampler/speculator setup, initial prompt prefill, and the speculative begin hook precede the timer; the result is not whole-request latency.

Calibration freezes R_{ref} as the median of prompt-median target latency per output token. A preregistered grid selects decision round and margin by exactness and calibration GM latency relative to the two-fixed-mode oracle, with frozen tie-breaks. The selected rule has margin 0, first decides after 8 rounds, then after each later completed round. Early-ending requests make no decision. Online control receives no task/family labels, future length, request-specific target latency or oracle outcomes.

Table 4. M4 ratios of paired prompt medians. Both speculative policies preserve 180/180 direct-ID sequences. Speedup and relative latency have opposite favorable directions. The oracle selects the lower of two measured fixed-mode medians with hindsight. Guard/Always is post-study secondary; values above one mean Guard is slower.

Ratio (×)	GM	95% prompt CI
Speedup: above one favors speculation		
Target / Guard	1.3114	[1.2487, 1.3787]
Target / Always-Spec	1.3469	[1.2867, 1.4088]
Relative latency: lower is better		
Guard / offline oracle	1.0278	[1.0137, 1.0446]
Post-study paired secondary latency ratio		
Guard / Always-Spec	1.0271	[1.0130, 1.0436]

6.2 Held-out evaluation

Historical outcomes informed the design; prospective means policy freeze before reserved outcomes. Exposure exclusion and deterministic selection give disjoint calibration, qualification and held-out splits of 40, 16 and 60 prompts, balanced across HumanEval and Spec-Bench. Policy and reference precede qualification/formal measurement; the appendix retains the audited freeze-hash metadata defect. Always-Spec and Guard each qualify 48/48 direct-ID pairs.

The prospective M4 study runs 3 repetitions of target-only, Always-Spec, and Guard on each held-out prompt, randomizing condition order within prompt and repetition: 540 observations. Both speculative conditions preserve 180/180 complete direct-ID sequences, with zero mismatches. For each prompt and condition, analysis takes the median latency across repetitions, then the geometric mean of prompt latency ratios. This ratio of condition medians differs from the historical M4 text-path hardware estimator. Confidence intervals use 10,000 prompt bootstrap resamples with seed 20260924; repetitions are not independent sampling units.

Guard is faster than target-only within the charged-loop boundary on this held-out workload (Table 4). Fixed speculation has the higher aggregate speedup point estimate. A reviewer-requested post-study paired secondary analysis of the frozen prompt medians gives Guard/Always-Spec latency 1.0271 [1.0130, 1.0436]. The interval is above parity: Guard had higher latency on this comparison. By prompt median, Guard is faster on 24 prompts, Always-Spec on 36, with 0 ties. Guard falls back in 30.6% of its 180 request executions; this measures control activity, not the fraction of beneficial decisions. Guard's geometric-mean latency relative to the offline oracle is 1.0278×. That oracle chooses the lower target-only or Always-Spec median after measurement and was unavailable to the runtime. It is a descriptive hindsight comparator, not an achievable-policy guarantee.

Reconstructing all original medians finds target-only faster than Always on only 2/60 prompts, both Spec-Bench. Always/oracle is 1.0007×. Choosing the better of these two measured fixed modes after seeing outcomes therefore offers about 0.069% GM latency reduction from Always, computed from the full-precision ratio. Guard falls back in all six executions of those two prompts but still loses to target-only on their medians. The family-stratified sensitivity analysis also leaves Guard slower (Appendix P). These post-study comparisons estimate neither transition cost nor opportunity for arbitrary within-request switching.

Guard preserves IDs but supplies no incremental benefit over Always-Spec. Fallback activity does not establish beneficial decisions or risk reduction. No runtime energy claim is made. Policy transfer and stable small-stratum effects remain unestablished; the archived policy-freeze defect and original stopped pipeline markers are documented separately.

7 System Implications and Research Outlook

Prediction equality, reference compatibility, intervention outcome and cost answer separate system questions. B570's exposed-cohort restoration and RTX's failed restoration candidate both admit successful prediction tests. Their different inputs and stage changes do not establish universal transfer or a shared numerical cause. Table 5 preserves the original qualification and cost decisions; those timing studies do not measure the new interventions.

Validate a candidate against its native path. The full-array predictions forecast restoration as well as persistent and new deviations. On RTX, the unchanged aggregate verdict conceals a restored task and a newly divergent task. A gate remains essential, but the paired path comparison explains why the frozen candidate must be rejected despite successful prediction. On B570, the candidate restores the tested exposed cohort; it is a controlled result rather than a production repair tool. Neither comparison establishes functional correctness, measured debugging savings or performance benefit.

Change the cost term that limits the opportunity. RTX serialization qualifies but loses within its instrumented boundary. On 358H, removing draft/residual cannot reach parity at fixed target work and progress. Under the optimistic premise that all other cost vanishes, a uniform target-work reduction of at least 16.28% CPU or 15.52% NPU is needed for the original prompt-GM ceiling to reach one. A design that changes target execution or committed progress must qualify that new path before claiming a gain. Isolated NPU speed cannot establish it.

Table 5. Engineering uses of the retained results. Reproduction concerns frozen outputs; optimization budgets are conditional and no proposed improvement is measured.

Study	Information beyond a basic gate	Reproduction task / optimization decision	Scope of that use
B570	Before/after predictions match; exposed diagnostic deviations restored	Retain target-only reproducer; accept restoration on this cohort for further qualification	Prefill plus sample/state handoff; no unique cause, cost or production claim
RTX intervention	Predictions match both paths, but one restoration cancels one new deviation	Reject the frozen recovery/no-regression candidate; assess each task's realized reference path	Different stage/input contract from B570; shared numerics; no transfer or cost claim
RTX B	Serialization qualifies but loses on every prompt	Exclude measured serialization as a latency optimization	Instrumented post-prefill cost; no qualified batched gain
358H	Overhead-only prompt-GM ceilings remain below one	Parity needs a changed premise; even with other cost zero, uniform target-work cuts must reach 16.28%/15.52%	Baseline/progress fixed; necessary optimistic budgets, not achieved savings
M4	About 0.069% two-fixed-mode opportunity; Guard adds 2.71% GM latency	Prefer fixed speculation on this aggregate measured workload; assess opportunity before control	Hindsight fixed modes; no bound on arbitrary within-request switching

Measure opportunity before adding control. Hindsight choice between the two measured fixed modes lowers GM latency from Always by about 0.069%; Guard instead adds 2.71%. This small fixed-mode opportunity helps explain why fallback activity alone is weak evidence of useful control. It does not bound policies that change the execution or switch within a request.

Stage attribution remains unresolved. A discriminating follow-up would need new frozen descriptions, qualifications and native comparisons for each changed stage, not post-hoc searching for an all-matching configuration. The present main claim is prediction under tested interventions and path-level reference effects; it does not require identifying a unique stage cause. Any further experiment should address a concrete competing explanation rather than enlarge the repeat count on already exposed tasks.

8 Related Work

Performance and heterogeneous execution. Speculative decoding amortizes target computation under its algorithmic assumptions (Leviathan et al., 2023). Prior work already documents verification and workload limits: Liu et al. (2026b) study verification dominance and batch effects in vLLM; Kim et al. (2026) account for conditional execution cost and runtime control; Lyu et al. (2026) derive edge–cloud profitability conditions. On consumer Apple silicon, Chordiya (2026) reports gains and slowdowns across five backend configurations, diagnoses ineffective batch-parallel verification in one quantized Metal path, tests output agreement, and observes target-only batch-shape argmax changes. Slowdown and correctness gating have already been studied, as has heterogeneous placement: Dovetail, DuoDecoding, and Mirror-SD study GPU/CPU or GPU/NPU paths (Zhang et al., 2025; Lv et al., 2025; Bhendawade et al., 2025); Ruiz y Mesa et al. (2026) combine an analytical cost model with compiler-oriented CPU/GPU partitioning, and Lever jointly optimizes draft, verification, and CPU/NPU execution for a smartphone target (Wang et al., 2026). State reuse has prior art in compressed draft KV memory (Yuan et al., 2026). Our 358H study measures one interface that combines replay and proposal; it establishes no general placement or persistence result.

Numerical paths and output contracts. The separate appendix compares the closest prior questions. Numerical sensitivity, deterministic inference construction, correctness forensics and cost gating are established prior work. LLM-42 uses fixed-shape verification and KV repair to construct deterministic inference (Gond et al., 2026); our tests forecast finite native paths and do not establish that guarantee. We test whether predefined programs forecast complete native outputs, including non-reference suffixes. The earlier tests select a simpler B570 reproducer and reject a specified RTX simplification; the intervention studies then check predictions against newly realized native paths. Separate gates and cost budgets constrain optimization. The new controlled studies predict native outputs before and after intervention: B570 restores its tested exposed cohort, while RTX fails its predefined restoration target. This is finite intervention validation, not a general correction method or

a unique-kernel diagnosis. Qin et al. (2026) formalize and verify fixed-deployment determinism through engine invariants and relational kernel contracts, including prefill/decode and KV-cache changes. Their bitwise-logit specification is stronger than our finite generated-ID comparisons. We forecast tested paths that can remain reference-incompatible; we supply neither an engine proof nor a deterministic implementation guarantee.

Adaptive speculation and the study boundary. TurboSpec adjusts speculation for serving goodput (Liu et al., 2025); BanditSpec selects configurations online (Hou et al., 2025); ACL HeteroSpec adapts verification from contextual entropy (Liu et al., 2026a). Adaptive control is established prior work. Our M4 Guard is a qualified runtime control test, not a new policy advantage: it beats target-only, but fixed speculation is faster in the post-study paired secondary comparison. Our measurements test prospective direct-ID qualification, frozen trajectory prediction, sampled intervention effects and realized cost on qualified paths.

9 Limitations

The tests use one narrow Qwen3 model pair, finite greedy endpoints and jointly varying hardware/backend/OS/quantization. Complete-output prediction covers generated IDs and endpoint only, not internal states or all execution events. No hardware ranking, functional pass@k, population failure rate or production guarantee follows. Reference compatibility is conditional on the specified reference, and historical text-path timing remains distinct from direct-ID qualification.

B570 restoration uses 24 already exposed tasks and includes first-pending sample/state handoff; it does not isolate prefill KV as a unique cause. Its earlier sampled fork changes scores without immediate splits, while delayed divergence is confounded by persistent continuation. RTX's 19 HumanEval and 24 MBPP tasks use a different intervention and proposal-conditioned description. No single predictor or intervention is shown to transfer across platforms. Shared numerical routines limit implementation independence; no proposal/history necessity, stage necessity, minimality or unique floating-point operation is established. Persistent mismatch is not automatically an output no-op. The RTX candidate's reference-recovery/no-new-regression target failed even though prediction/native equality passed.

Exposure audits cover retained project records, not model training. Three repetitions are stability observations; positions within trajectories and repeated arrays are correlated. Hashes bind content, while reported host/server order and Git records do not authenticate absolute experimental time. RTX remote readback is stronger ordering evidence than B570 local seals, but neither certifies an independent external experiment clock. Offline raw review does not execute models or authenticate omitted weights/binaries. Checked KV/logits and sampler-clone behavior do not establish byte-equal opaque sampler state.

Intervention studies provide no latency or energy result. Historical RTX cost changes multiple stages, uses effective settings distinct from the new candidate, and measures instrumented post-prefill latency including logging. Its qualified serialized path is slower, with no deployment, throughput or energy estimate. The 358H region combines reconstruction/replay with proposal; separate costs and some runtime raw traces are unavailable. Conditional ceilings fix measured baseline, target work and committed progress; uniform-reduction budgets assume all other costs vanish and are not achieved improvements. Blocked persistence/wider-target paths establish no architectural impossibility.

M4 Guard/Always, family sensitivity and fixed-mode opportunity are post-study analyses. Its oracle chooses two measured fixed modes with hindsight, not arbitrary within-request policies. Confidence intervals condition on selected prompt rows, observed medians and configurations and exclude session/recalibration uncertainty. The metadata defect and original stopped markers remain documented. Runtime energy is unmeasured, energy domains differ, and the historical corrected synthetic ceiling was unexecuted.

10 Conclusion

Under the tested fixed stacks, frozen execution-path descriptions predict complete native generated IDs and endpoints before and after two controlled interventions. B570 restores reference agreement on its exposed diagnostic/control cohort; RTX predicts the changed paths but fails its predefined recovery and no-new-regression target. Equal RTX aggregate agreement conceals one restoration and one new deviation. The studies validate distinct descriptions and configurations, not a universal repair or stage cause. Qualification and cost then support separate decisions: historical RTX serialization is compatible but slower, 358H overhead-only ceilings remain below parity, and M4 Guard loses to fixed speculation. Predict the realized path, evaluate its reference compatibility, and measure its actual cost before calling it an optimization.

REFERENCES

Bhendawade, N., Nishu, K., Kundu, A., Bartels, C., Cho, M., and Belousova, I. Mirror speculative decoding: Breaking the serial barrier in LLM inference. *arXiv preprint arXiv:2510.13161v2*, 2025. URL https://arxiv.org/abs/2510.13161v2.

Chen, M., Tworek, J., Jun, H., Yuan, Q., Ponde de Oliveira Pinto, H., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., Sastry, G., Mishkin, P., Chan, B., Gray, S., Ryder, N., Pavlov, M., Power, A., Kaiser, L., Bavarian, M., Winter, C., Tillet, P., Petroski Such, F., Cummings, D., Plappert, M., Chantzis, F., Barnes, E., Herbert-Voss, A., Hebgen Guss, W., Nichol, A., Paino, A., Tezak, N., Tang, J., Babuschkin, I., Balaji, S., Jain, S., Saunders, W., Hesse, C., Carr, A. N., Leike, J., Achiam, J., Misra, V., Morikawa, E., Radford, A., Knight, M., Brundage, M., Murati, M., Mayer, K., Welinder, P., McGrew, B., Amodei, D., McCandlish, S., Sutskever, I., and Zaremba, W. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374v2*, 2021. URL https://arxiv.org/abs/2107.03374v2.

Chordiya, P. Lossless but not free: An empirical anatomy of speculative decoding on consumer hardware. *arXiv preprint arXiv:2607.17283*, 2026. doi: 10.48550/arXiv.2607.17283. URL https://arxiv.org/abs/2607.17283.

Gond, R., Kamath, A. K., Ramjee, R., and Panwar, A. LLM-42: Enabling determinism in LLM inference with verified speculation. *arXiv preprint arXiv:2601.17768v2*, 2026. URL https://arxiv.org/abs/2601.17768v2.

Hou, Y., Zhang, F., Du, C., Zhang, X., Pan, J., Pang, T., Du, C., Tan, V., and Yang, Z. BanditSpec: Adaptive speculative decoding via bandit algorithms. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 24045–24079. PMLR, 2025. URL https://proceedings.mlr.press/v267/hou25h.html.

Kim, S., Kim, J., Cho, Y., and Seo, J. Runtime-controlled evaluation of speculative decoding methods. *Electronics*, 15(14):3179, 2026. doi: 10.3390/electronics15143179. URL https://www.mdpi.com/2079-9292/15/14/3179.

Leviathan, Y., Kalman, M., and Matias, Y. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pp. 19274–19286. PMLR, 2023. URL https://proceedings.mlr.press/v202/leviathan23a.html.

Liu, S., Ye, Y., Zhu, Q., Cao, Z., and He, Y. HeteroSpec: Leveraging contextual heterogeneity for efficient speculative decoding. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 12930–12947. Association for Computational Linguistics, July 2026a. doi: 10.18653/v1/2026.acl-long.589. URL https://aclanthology.org/2026.acl-long.589/.

Liu, X., Park, J., Hu, L., Kwon, W., Li, Z., Zhang, C., Du, K., Mo, X., You, K., Cheung, A., Deng, Z., Stoica, I., and Zhang, H. TurboSpec: Closed-loop speculation control system for optimizing LLM serving goodput. *arXiv preprint arXiv:2406.14066v3*, 2025. URL https://arxiv.org/abs/2406.14066v3.

Liu, X., Yu, J., Park, J., Stoica, I., and Cheung, A. Speculative decoding: Performance or illusion? In *Proceedings of Machine Learning and Systems*, volume 8, pp. 1719–1741, 2026b. URL https://proceedings.mlsys.org/paper_files/paper/2026/hash/554e056fe2b6d9fd27ffcd3367ae1267-Abstract-Conference.html.

Lv, K., Guo, H., Guo, Q., and Qiu, X. DuoDecoding: Hardware-aware heterogeneous speculative decoding with dynamic multi-sequence drafting. *arXiv preprint arXiv:2503.00784*, 2025. URL https://arxiv.org/abs/2503.00784.

Lyu, Y., Irukulapati, B., and Champati, J. P. Speculation at a distance: Where edge-cloud speculative decoding actually pays off. *arXiv preprint arXiv:2606.25091v2*, 2026. URL https://arxiv.org/abs/2606.25091v2.

Qin, J., Du, A., Zhang, D., Lentz, M., and Zhuo, D. Vosti: Specifying, implementing, and verifying deterministic llm inference. *arXiv preprint arXiv:2609.38981v1*, 2026. URL https://arxiv.org/abs/2609.38981v1.

Ruiz y Mesa, A., Korol, G., Riesterer, M., Cardoso de Lima, J. P., and Castrillon, J. Compiler-assisted speculative sampling for accelerated LLM inference on heterogeneous edge devices. In *8th Workshop on Accelerated Machine Learning (AccML), co-located with HiPEAC*, January 2026. URL https://accml.dcs.gla.ac.uk/papers/2026/8th_AccML_paper_8.pdf.

Wang, T., Li, F., Sun, Y., Gao, W., and Ren, J. Lever: Speculative LLM inference on smartphones. *arXiv preprint arXiv:2605.16786*, 2026. doi: 10.48550/arXiv.2605.16786. URL https://arxiv.org/abs/2605.16786.

Xia, H., Yang, Z., Dong, Q., Wang, P., Li, Y., Ge, T., Liu, T., Li, W., and Sui, Z. Unlocking efficiency in large language model inference: A comprehensive survey of speculative decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 7655–7671. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.456. URL https://aclanthology.org/2024.findings-acl.456/.

Yuan, J., Li, H., Ding, X., Xie, W., Li, Y.-J., Zhao, W., Wan, K., Shi, J., Hu, X., and Liu, Z. Understanding and mitigating numerical sources of nondeterminism in LLM inference. *arXiv preprint arXiv:2506.09501v2*, 2025. URL https://arxiv.org/abs/2506.09501v2.

Yuan, T., Liao, C., and Wen, Z. Strong drafts need compact memories: Long-context speculative decoding with compressed KV cache. *arXiv preprint arXiv:2608.30252*, 2026. URL https://arxiv.org/abs/2608.30252. EMNLP 2026 Findings.

Zhang, L., Zhang, Z., Xu, B., Li, R., Tian, Z., Mei, S., and Li, D. Dovetail: A CPU/GPU heterogeneous speculative decoding for LLM inference. *arXiv preprint arXiv:2412.18934v2*, 2025. URL https://arxiv.org/abs/2412.18934v2.

Zhang, R. H., Dey, S., Mishra, A., Wu, H., Li, B., and Zhang, R. Correctness forensics for batch speculative decoding: Diagnosing the ragged tensor problem. *arXiv preprint arXiv:2510.22876v4*, 2026. URL https://arxiv.org/abs/2510.22876v4. Accepted to Findings of EMNLP 2026.